

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, understanding language, and recognizing patterns.

Machine Learning (ML) is a subset of Artificial Intelligence. It allows computers to learn from data and improve their performance without being explicitly programmed for every task. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Python is one of the most popular programming languages for AI and machine learning. Libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch are widely used to build AI applications.

Retrieval Augmented Generation (RAG) is a technique that combines information retrieval with generative AI. A RAG system first retrieves relevant information from a knowledge base and then uses that information to generate a useful answer.